

- Midterm Defense Q&A Cheat Sheet
 - 1. Data & Generalization (数据与泛化能力)
 - 2. Model Selection (模型演进与选择)
 - 3. Performance Metrics (性能指标解释)
 - 4. SaaS Architecture (SaaS 架构与高并发)
 - 5. Project Scope & Priorities (项目规划与工作量)

Midterm Defense Q&A Cheat Sheet

1. Data & Generalization (数据与泛化能力)

Q (EN): How can you claim your model is reliable with an AP50 of 1.0 when your bird nest dataset only has 54 images? Won't this overfit to field data? **Q (CN):** 你的鸟巢数据集只有54张图片，却达到了 AP50=1.0。你怎么证明这不是过拟合？到了实际电网巡检中能用吗？

A (EN):

- Thank you for the question. We acknowledge that 1.0 AP50 on a same-distribution validation set does not guarantee field performance across new backgrounds or weather conditions.
- However, we actively prevented standard overfitting during training by using a highly **deterministic augmentation pipeline**, specifically our custom "Random IoU Crop", which forces the model to learn from partial, multi-scale object features rather than memorizing entire images. We also avoided generative AI augmentation because it introduced physically implausible artifacts.
- In the next phase, the platform's priority is testing against cross-scene hard samples (e.g., haze, backlight) to validate and improve real-world generalization.

A (CN):

- 感谢老师的提问。我们确实承认，在同分布验证集上达到 1.0 的 AP 并不能代表它能在未见过的野外背景（如雾霾逆光等）下一样完美。

- 但是，我们在训练时非常注意防止死记硬背式的过拟合。我们的策略是使用了**确定性的强数据增强**，特别是自定义的 Random IoU Crop，强迫模型从局部、多尺度的特征中学习。我们也没有用生成式增强，因为早期实验发现生成式方法会引入物理上不合理的伪影，反而干扰学习。
 - 下一阶段，平台的重心就是引入跨场景的困难负样本进行平台级的泛化测试。
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2. Model Selection (模型演进与选择)

Q (EN): Why are you using three completely different technical approaches (Mask R-CNN, YOLO, OCR) for the three modules? Why not unify them into one large model?

Q (CN): 为什么你的三个模块用了三种截然不同的技术方案（实例分割、目标检测、纯算法 OCR）？为什么不用一个统一的大模型来做？

A (EN):

- The technical choices perfectly align with the specific **business requirements** of each defect type.
- Bird nests require precise boundary calculations to determine the exact hazard zone for work-order dispatches, hence the need for instance segmentation (**Mask R-CNN**).
- Oil leaks simply require the system to trigger a "presence and approximate location" alert with high speed, making object detection (**YOLOv8**) optimal.
- Infrared images already have standardized temperature scales. Using robust **OCR and pixel-intensity mapping** solves the problem precisely without the overhead of labeling and training a heavy deep-learning model from scratch.

A (CN):

- 这样的技术选型是完全由电网的**具体业务需求**驱动的。
 - 鸟巢检测需要精确的边界信息，从而在派发工单时计算安全净距，所以必须用实例分割（**Mask R-CNN**）。
 - 漏油检测的核心诉求是快速报警，只需要知道“有没有、大概在哪”，因此选用速度占优的目标检测（**YOLOv8**）。
 - 红外测温因为图像本身带有规范的温度刻度条，使用纯视觉算法（**OCR+像素映射**）既准确又轻量，完全避免了从零标注和训练深度学习模型的巨大成本。
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3. Performance Metrics (性能指标解释)

Q (EN): Your oil leak recall is only 0.34, with an mAP50 around 0.39. Is this module actually usable in a production environment? **Q (CN):** 你的漏油模块 Recall 只有 0.34 , mAP50 也不到 0.4。这个指标拿得出手么？能在实际中用么？

A (EN):

- The current metrics are from "train10", which was trained **only** on the CSDN dataset subset as an initial integration test for the Flask pipeline.
- While the recall is low (0.34), the precision is relatively high (0.83). This indicates the current model is highly **conservative**—it misses some leaks but produces very few false alarms, which is actually preferable to alert fatigue in a real-world warning system.
- The immediate next step in Week 11 is to retrain the module on our newly unified VOC+CSDN merged dataset, which we anticipate will significantly raise both the recall and the overall mAP.

A (CN):

- 目前在中期报告里展示的这组数据，是完全基于 CSDN 子集训练的初步结果，目的是为了跑通打通后端的 Flask 推理全流程。
- 尽管召回率偏低（0.34），但精确率相对比较高（0.83）。这说明当前模型更倾向于“保守报警”——宁可漏报，也不瞎报。在真实的电网告警系统中，为了避免运维人员产生“告警疲劳”，高 Precision 往往比高 Recall 更重要一点。
- 下一阶段立马要做的就是使用我们刚刚清洗合并的 VOC+CSDN 完整数据集重新训练，届时 Recall 和 mAP 都会有显著提升。

4. SaaS Architecture (SaaS 架构与高并发)

Q (EN): You mentioned this is a SaaS backend. Flask is synchronous by default. How does your platform handle real-world concurrent high loads from multiple drones? **Q (CN):** 你们做的是 SaaS 平台。但 Flask 默认是单线程同步的。如果有几百架无人机同时回传图片，你的后端怎么处理高并发？

A (EN):

- Currently, our production deployment uses **Gunicorn with multiple worker processes**, which handles moderate concurrency well enough for the midterm prototype demonstration.
- However, we are fully aware that true enterprise scalability requires decoupling the API from the inference engines.
- Setting up an **asynchronous task queue** (like Celery with Redis) to handle off-loaded batch inference tasks is exactly the core objective of "Track 1" in our next phase's work plan.

A (CN):

- 目前我们的中期原型是用 **Gunicorn 开启多进程 Worker** 来部署的，对于目前的演示和轻量级并发是完全够用的。
- 但我们也深知，真正企业级的高并发 SaaS 必须把 API 接口层和重计算的推理层解耦。
- 所以在我们下一阶段的 "Track 1" 规划中，核心任务之一就是引入**异步任务队列**（比如 Celery配合消息中间件），前端只管上传，后端从队列抓取图片异步执行推理。

5. Project Scope & Priorities (项目规划与工作量)

Q (EN): You have only a few weeks left, and you need to improve model accuracy while also building SaaS features like RBAC. What is your actual priority? **Q (CN):** 离最终答辩没多少周了，你既要优化模型精度（做泛化实验），又要搞 SaaS 平台的权限管控。你一个人做的完吗？你的优先级是什么？

A (EN):

- My core strategy has always been "**Vertical Slice First, then Horizontal Scale**".
- Therefore, my absolute priority is delivering a demonstrable, multi-model prototype that proves the complete AI vision pipeline works flawlessly end-to-end.
- Improving cross-scene generalizability of the models takes precedence. Deep enterprise capabilities, such as complex OAuth2 or granular RBAC, are considered "nice-to-have" and will only be implemented if time permits.

A (CN):

- 我贯穿项目的核心策略是**“先打通垂直切片，再做横向扩展”**。
- 因此，我目前的绝对优先级是拿出一个“演示效果好、多模型可编排调度”的完整 AI 闭环系统，打通所有的底层技术路径。
- 在接下来的工作里，我会把精力优先放在提高各个模型模块在实际电网场景的泛化能力上。至于 OAuth2 或者非常细粒度的 RBAC 权限管控，属于锦上添花的 SaaS 商业功能，如果有时间充足我才会去完善。